ME 321 Mechanical Engineering Analysis for Design

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Pump Assembly Final Report

Group Number: **3**

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Contents

1 Introduction **3**

2 Equilibrium Calculations **3**

2.1 Free Body Diagrams . . . 3

2.1.1 Support A . . . 4

2.1.2 Support B . . . 4

2.1.3 Beam A . . . 5

2.1.4 Beam B . . . 6

2.1.5 Plate . . . 6

2.1.6 Assembly . . . 7

2.2 Power and Torque Calculations . . . 7

2.3 Internal Load Diagrams . . . 8

2.3.1 Beam B . . . 8

2.3.2 Support B . . . 10

2.3.3 Pump Plate . . . 10

3 Stress Analysis (Symbolic) **11**

3.1 Combined Loading Modes and Stress Types . . . 11

3.2 Stress Concentrations . . . 12

3.3 Critical Locations . . . 12

3.4 Nominal Stress Expressions . . . 13

3.5 Maximum Shear and Principal with Mohr's Circle . . . 15

4 Failure Determination (Symbolic) **18**

4.1 Factor of Safety Expressions . . . 18

4.2 Maximum Deflection of Pump Platform . . . 19

4.3 Critical Buckling Load of Support Arms . . . 20

5 Design Iteration **20**

5.1 Initial Dimensions . . . 20

5.2 First Design Iteration . . . 21

5.3 Ansys Material Selection Plots . . . 23
5.4 Further Iterations . . . 25
5.5 Final Dimensions and Materials . . . 26

6 Finite Element Validation 26
6.1 FEA Setup . . . 26
6.2 CAD Assembly . . . 28
6.3 FEA Results . . . 29
6.4 Convergence Study . . . 31
6.5 GA3 Questions . . . 32

7 Design Changes 33

8 CAD and Fabrication Drawings 33
8.1 CAD Drawings . . . 33
8.2 Bill of Materials . . . 37
8.3 Physical Component Photographs . . . 37

9 Physical Prototype Performance Results 37

10 Discussion 38

11 Conclusion 39

A Detailed Stress Derivations 41
A.1 Beam B – Point F . . . 41
A.2 Principal Stress Calculation . . . 42
A.3 Pump Plate Deflection . . . 42
A.4 Expanded Principal Stress Calculations . . . 42
A.5 Pump Plate Expansion . . . 43
A.6 Pin Stress Expansion . . . 43

B Detailed Deflection Derivation 43

C Detailed Buckling Analysis 45

1. Introduction

Following flooding events, effective water removal from low-lying areas is critical for cleanup operations. PumpCo has proposed a small-capacity centrifugal pump system in which the pump head is mounted on a structural support frame and driven by a belt and pulley system. The objective of this project was to design, analyze, and fabricate a pump support structure capable of securely mounting the pump while withstanding operational loads.

The support structure was required to interface with an existing frame using removable clevis pins and include at least two support arms to counteract torsional effects from the pulley system. Key design constraints included a maximum height of 61 cm, a pump mass of 5.67 kg, specified mounting hole geometry, and belt forces with a minimum tension of 200 N. Additionally, the structure was required to limit deflection at the end of the platform to less than 1 mm and maintain a minimum factor of safety of 1.5 against yielding and buckling. Material selection was constrained by corrosion resistance requirements and a total budget of $150.

The general approach taken by the team involved first performing equilibrium and stress analyses to determine internal forces and identify critical loading conditions. These results were used to develop symbolic expressions for stress, deflection, and failure criteria, which were then evaluated numerically during the design iteration process. Material selection was done using Ashby charts to balance strength, stiffness, and weight while meeting cost and manufacturability constraints. The final design was validated using finite element analysis and tested through fabrication of a physical prototype.

This report is organized as follows: Section 2 presents equilibrium calculations used to determine system forces and reactions. Section 3 outlines the stress analysis, including identification of loading modes and critical locations. Section 4 details failure criteria and safety considerations. Section 5 describes the design iteration process and final design selection. Section 6 presents finite element validation results. Section 7 discusses design changes made throughout the project. Section 8 includes CAD drawings and fabrication details. Section 9 summarizes physical prototype performance, and Section 10 provides discussion of results and observed discrepancies.

2. Equilibrium Calculations

2.1 Free Body Diagrams

A free body diagram (FBD) was generated for the whole assembly and for each unique component. Supports A and B and Beams A and B shared FBDs, where force vectors were labelled with forces for either component. Reference Figure 1 for naming conventions and key point locations.

Geometric parameters and applied loads referenced throughout the equilibrium equations are summarized in Table 1. By symmetry of the wall mount, the vertical reactions at pins C and D are equal ($C_z = D_z$), which closes the otherwise indeterminate system.

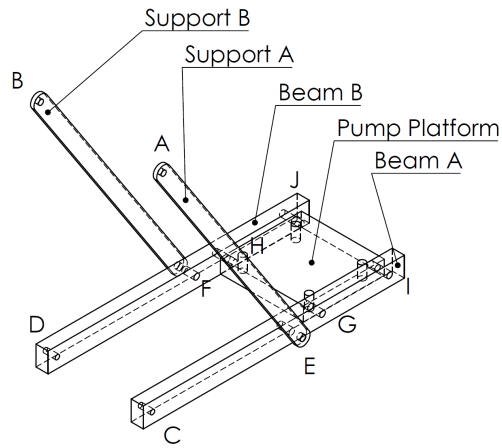


Figure 1: Component names and key points in the assembly

Symbol	Value	Symbol	Value
z_s	0.3048 m	P_z	550 N
x_s	−0.2441 m	M_{Px}	−4 N m
x_{b1}	−0.2441 m	M_{Py}	15 N m
x_{b2}	−0.3329 m	x_P	−0.2885 m
y_{AB}	0.082 55 m	W_{pump}	55.62 N

Table 1: Geometric parameters and applied loads.

2.1.1 Support A

Reference Figure 2 for the FBD for this component. Note that force vectors E and F have x and z components E_x, E_z, F_x and F_z respectively.

Force Balance

$$\sum F_x = 0 : \quad A_x + E_x = 0 \tag{1}$$
$$\sum F_z = 0 : \quad A_z + E_z = 0 \tag{2}$$

Moment Balance

$$\sum M_y^A = 0 : \quad -E_x z_s - E_z x_s = 0 \tag{3}$$

2.1.2 Support B

Refer to Figure 2 for the FBD for this component.

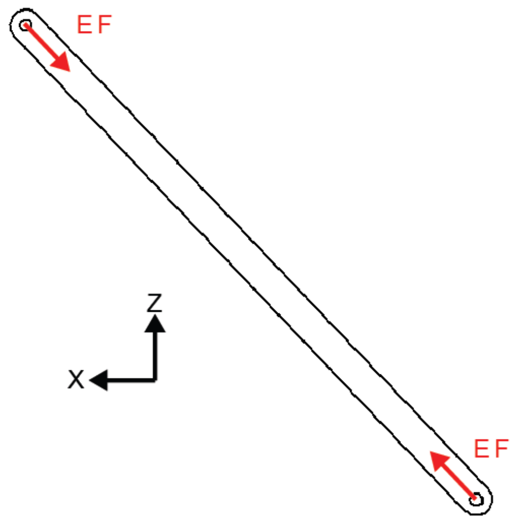


Figure 2: FBD for Supports A and B

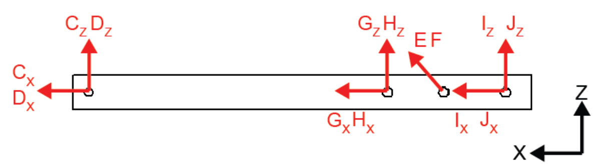


Figure 3: FBD for Beam A and Beam B

Force Balance

$$\sum F_x = 0 : \quad B_x + F_x = 0 \tag{4}$$
$$\sum F_z = 0 : \quad B_z + F_z = 0 \tag{5}$$

Moment Balance

$$\sum M_y^B = 0 : \quad -F_x z_s - F_z x_s = 0 \tag{6}$$

2.1.3 Beam A

Force Balance

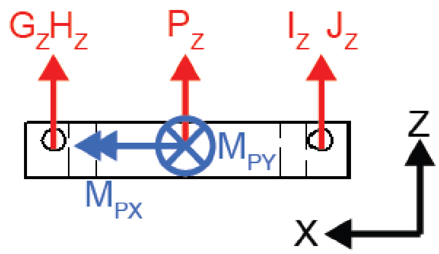


Figure 4: FBD for Pump Plate

Reference Figure 3 for the FBD for this component.

$$\sum F_x = 0: \quad C_x - E_x = 0 \tag{7}$$
$$\sum F_z = 0: \quad C_z - E_z + G_z + I_z = 0 \tag{8}$$

Moment Balance

$$\sum M_y^C = 0: \quad x_s E_z - x_{b1} G_z - x_{b2} I_z = 0 \tag{9}$$

2.1.4 Beam B

Reference Figure 3 for the FBD for this component.

Force Balance

$$\sum F_x = 0: \quad D_x - F_x = 0 \tag{10}$$
$$\sum F_z = 0: \quad D_z - F_z + H_z + J_z = 0 \tag{11}$$

Moment Balance

$$\sum M_y^D = 0: \quad x_s F_z - x_{b1} H_z - x_{b2} J_z = 0 \tag{12}$$

2.1.5 Plate

Reference Figure 4 for the FBD for this component.

Force Balance

$$\sum F_z = 0: \quad P_z - G_z - H_z - I_z - J_z - W_{\text{pump}} = 0 \tag{13}$$

Moment Balance

$$\sum M_x = 0: \quad y_{AB}(-G_z + H_z - I_z + J_z) + M_{Px} = 0 \tag{14}$$

$$\sum M_y = 0: \quad x_{b1}(G_z + H_z) + x_{b2}(I_z + J_z) - x_P(P_z - W_{\text{pump}}) + M_{Py} = 0 \tag{15}$$

2.1.6 Assembly

Equations (1) through (15), together with the symmetry constraint $C_z = D_z$, form a determinate system of 16 unknowns. The system was solved using a SymPy-based solver, and all equilibrium summations were verified to be less than 10^{-6} N or N·m. These equations were all entirely symbolic, so further design iteration was simply a matter of adjusting parameters. The resulting reaction values are summarized in Table 2 for the GA2 design.

Reaction	Value (N)	Reaction	Value (N)	Reaction	Value (N)	Reaction	Value (N)
A_x	−101.52	B_x	−126.95	C_x	+101.52	D_x	+126.95
A_z	−144.34	B_z	−192.79	C_z	+187.87	D_z	+187.87
E_x	+101.52	F_x	+126.95	G_z	−214.13	H_z	+16.03
E_z	+242.44	F_z	+290.89	I_z	+366.80	J_z	+185.10

Table 2: Solved reaction forces for all pins and bolts.

2.2 Power and Torque Calculations

The pump operates on water ($\rho = 1000\,\text{kg}\,\text{m}^{-3}$) at a volumetric flow rate of $Q = 0.009\,464\,\text{m}^3\,\text{s}^{-1}$ with a pump efficiency of $\eta = 0.65$. The static head is

$$h_s = Z_2 - Z_1 = 18.29\,\text{m} \tag{1}$$

Neglecting minor losses, the total dynamic head equals $H = h_s$. The hydraulic power delivered to the fluid is

$$P_h = \rho g Q H = (1000)(9.81)(0.009464)(18.29) \approx 1696\,\text{W} \tag{2}$$

Accounting for pump efficiency, the required shaft power is

$$P_{\text{shaft}} = \frac{\rho g Q H}{\eta} = \frac{1696}{0.65} = 2612\,\text{W} \tag{3}$$

At the rated speed of 1800 RPM, the shaft angular velocity is

$$\omega = \frac{2\pi \cdot 1800}{60} = 188.5\,\mathrm{rad\,s^{-1}} \tag{4}$$

The shaft torque is

$$T = \frac{P_{\mathrm{shaft}}}{\omega} = \frac{2612}{188.5} = 13.86\,\mathrm{N\,m} \tag{5}$$

The torque is transmitted through a belt drive. With a pulley radius of $r = 0.127\,\mathrm{m}$ and a slack-side belt tension of $F_2 = 200\,\mathrm{N}$, the tight-side tension F_1 is found from the torque equation:

$$T = (F_1 - F_2)\,r \tag{6}$$

$$13.86 = (F_1 - 200)(0.127) \tag{7}$$

$$F_1 = \frac{13.86}{0.127} + 200 = 309.13\,\mathrm{N} \tag{8}$$

The net belt force acting laterally on the pulley is

$$F_{\mathrm{belt}} = F_1 + F_2 = 309.13 + 200 = 509.13\,\mathrm{N} \tag{9}$$

This lateral force, transmitted through the pulley at a radius $r = 0.127\,\mathrm{m}$ from the shaft centerline, produces a resultant moment on the pump mount of

$$M_{\mathrm{pulley}} = F_{\mathrm{belt}} \cdot r = (509.13)(0.127) = 64.66\,\mathrm{N\,m} \tag{10}$$

which is reacted by the mounting bolts and carried into the plate as the applied moment M_{Py} in the plate equilibrium.

2.3 Internal Load Diagrams

Because of the identical geometry and similar loading of each pair of components, internal force diagrams were created for the more critical beam and support. Each diagram shows shear, bending moment, torque, and axial load along the member span.

2.3.1 Beam B

See Figure 5 for internal forces in the X direction for beam B. The maximum bending moment occurs at the support pin connection ($x = -0.2\,\mathrm{m}$), where $M = 37.57\,\mathrm{N\,m}$ and $V = -89.76\,\mathrm{N}$. This location is the overall most critical point in the assembly, and the adjacent pin hole introduces an additional stress concentration that is addressed in Section 3.

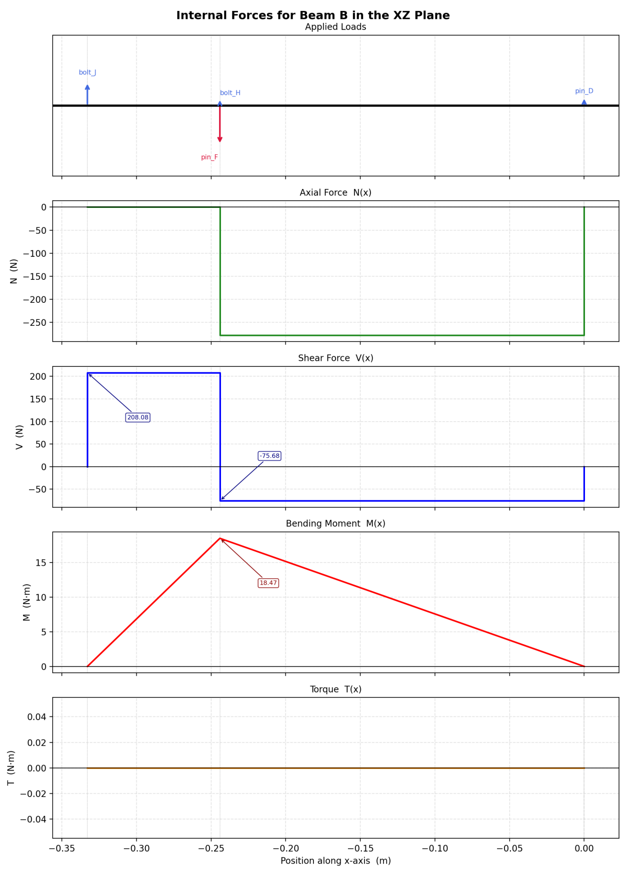


Figure 5: Internal forces for beam B along the X axis

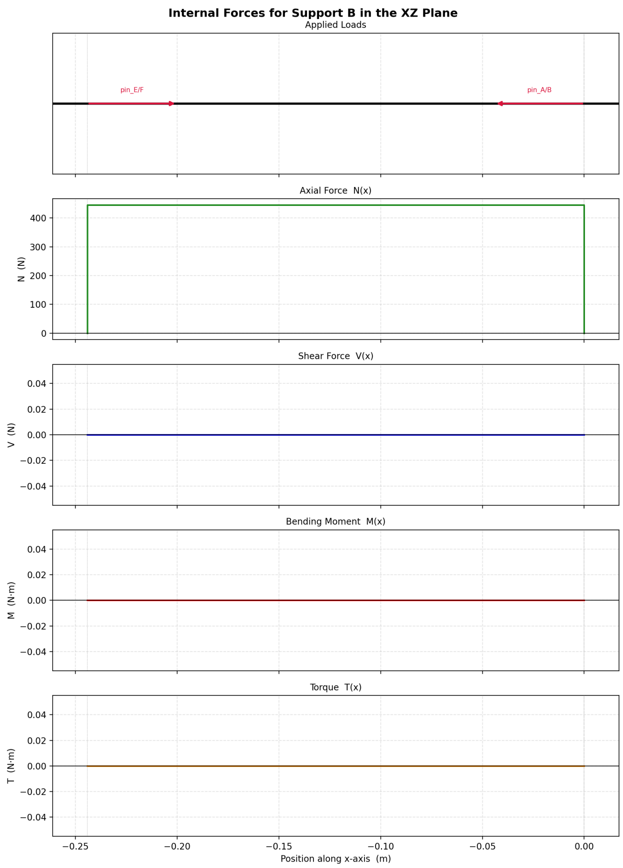


Figure 6: Internal forces for Support B along the support beam's length

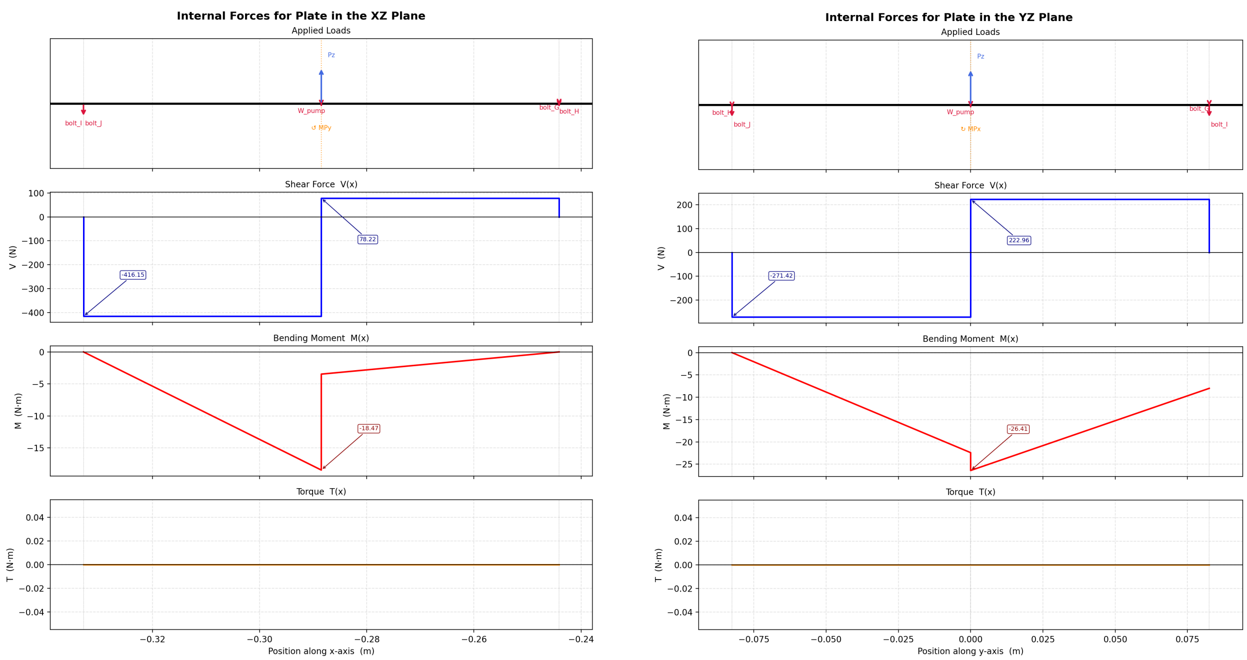
2.3.2 Support B

Figure 6 shows internal forces for support B, which is under more compression than support A. As a two-force member, the support carries only an axial compressive load of 317.39 N, with negligible shear, bending, and torsion along its length.

2.3.3 Pump Plate

The pump plate experiences forces along the XZ plane and the YZ plane. Therefore, two internal force diagrams are shown.

In the XZ plane (Figure 7a), the maximum shear of $-551.9\,\mathrm{N}$ occurs at the bolt row closest to the pump load ($x = -0.39\,\mathrm{m}$), together with a constant torque of $-4\,\mathrm{N\,m}$ transmitted through the plate from the applied moment M_{Px}.



(a) XZ plane **(b)** YZ plane

Figure 7: Internal forces in the pump plate

In the YZ plane (Figure 7b), the maximum bending moment of $-20.60\,\mathrm{N\,m}$ occurs at the plate centerline ($y = 0$), where the lateral offset between the beam attachment rows produces peak bending about the x-axis.

3. Stress Analysis (Symbolic)

3.1 Combined Loading Modes and Stress Types

Based on the free body diagrams and loading conditions, each component in the assembly experiences a combination of loading modes. The primary stress types for each component are summarized in Table 3.

Table 3: Stress types experienced by each component in the assembly.

Component	Tensile	Compressive	Bending	Trans. Shear	Dir. Shear	Torsion
Beam A			✓	✓		
Beam B			✓	✓		
Support A		✓				
Support B		✓				
Pump Plate			✓	✓		
Pins and Bolts					✓	

The beams experience combined bending and transverse shear due to applied loads and support reactions. The support members are primarily in compression, making them susceptible to buckling. The pump plate undergoes bending and shear due to the distributed load from the pump, while the pins and bolts primarily experience direct shear at connection points.

3.2 Stress Concentrations

Stress concentrations occur at geometric discontinuities such as holes, fillets, and connection points, where local stresses can exceed nominal values. In this design, these regions were identified as potential locations of elevated stress; however, stress concentration factors were not explicitly determined from standard charts, as only ductile materials were used.

The primary stress concentration locations in the design are summarized below:

- **Beam A:** Pin holes at points C, E, G, and I; pin pullout regions at C and E.
- **Beam B:** Pin holes at points D, F, H, and J; pin pullout regions at D and F.
- **Support A:** Pin holes at A and E.
- **Support B:** Pin holes at B and F.
- **Pump Plate:** Bolt holes at G, H, I, and J; local stress increases due to geometric discontinuities near these connections.

Stress concentration factors were not explicitly applied in this analysis. Since all selected materials are ductile, the design was evaluated using the Von Mises (distortion energy) criterion, which accounts for the combined stress state rather than relying on peak local stresses. While geometric discontinuities such as holes and connections still introduce localized stress increases, their effects are partially mitigated by the ability of ductile materials to yield and redistribute stress. As a result, nominal stress expressions were considered sufficient for predicting overall component performance.

3.3 Critical Locations

Based on the identified loading conditions and stress concentrations, several critical locations were determined where failure is most likely to initiate. These locations correspond to regions of maximum combined stress and geometric discontinuities.

- **Beam A:** The region near the pin hole at point E, particularly along the top edge, where bending stresses are highest and amplified by the stress concentration from the hole.
- **Beam B:** The region near the pin hole at point F, specifically along the top edge, due to the combination of maximum bending stress and stress concentration effects.

- **Support Arms (A & B):** The midpoint of each support arm, where compressive loading makes the members most susceptible to buckling.
- **Pump Plate:** The regions around the mounting holes at points G, H, I, and J, where bending stresses combine with stress concentrations from the bolt holes.
- **Pins (e.g., Pin F):** The shear plane within the pin, where direct shear stresses are highest due to load transfer between connected members.

These critical locations are used in subsequent analysis to evaluate maximum stresses, factor of safety, and potential failure modes.

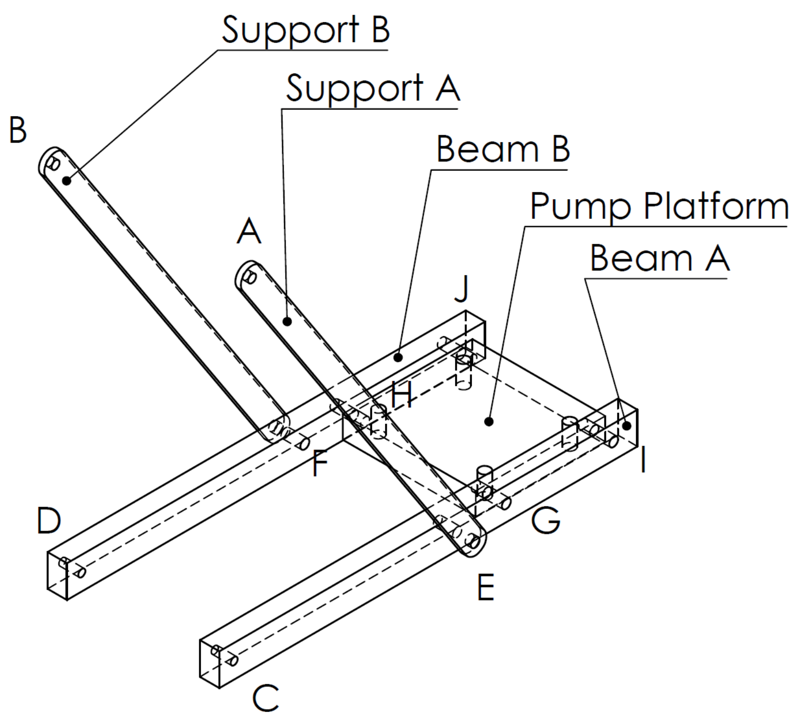


Figure 8: Diagram indicating potential critical locations. All dimensions are left as symbolic variables.

3.4 Nominal Stress Expressions

Nominal stress expressions were developed for each critical component using the geometry and loading conditions identified from equilibrium analysis. These expressions incorporate the specific cross-sectional properties of each member and serve as the basis for evaluating maximum stresses.

Beam B – Critical Location (Point F):

The beam experiences combined bending, axial loading, and transverse shear. For a hollow rectangular cross-section:

$$\sigma_{\text{bending}} = \frac{Mc}{I} = \frac{\left(\frac{PL_{\text{beam}}}{2}\right)\left(\frac{W_o}{2}\right)}{\frac{W_o^4 - w_i^4}{12}} = \frac{3PL_{\text{beam}}W_o}{W_o^4 - w_i^4} \tag{11}$$

$$\tau_{\text{transverse}} = \frac{2V}{A} = \frac{2P}{W_o^2} \tag{12}$$

Support Arm – Midpoint:

The support arms are subjected to axial compression:

$$\sigma_{\text{axial}} = \frac{P}{w_{\text{support}}h_{\text{support}}} \tag{13}$$

Pin (e.g., Pin F):

Pins experience direct shear:

$$\tau_{\text{direct}} = \frac{V}{A} = \frac{4P}{\pi d_{\text{pin}}^2} \tag{14}$$

Pump Plate – Mid-Span:

The pump plate experiences bending and shear:

$$\sigma_{\text{bending}} = \frac{\left(\frac{P_pL_p}{2}\right)\left(\frac{h_p}{2}\right)}{\frac{W_ph_p^3}{12}} = \frac{3P_pL_p}{W_ph_p^2} \tag{15}$$

$$\tau_{\text{transverse}} = \frac{2P}{W_ph_p} \tag{16}$$

These nominal stresses are later combined and modified using stress concentration factors to determine the maximum stresses at each critical location. Full derivations and intermediate steps are provided in Appendix A.

3.5 Maximum Shear and Principal with Mohr's Circle

At each critical location, the combined loading conditions result in multi-axial stress states. Principal stresses and maximum shear stresses were determined using Mohr's circle relationships.

Beam B – Critical Location (Point F):

The beam experiences combined bending and transverse shear. Assuming plane stress:

$$\sigma_x = \sigma_{\text{bending}}, \quad \sigma_y = 0, \quad \tau_{xy} = \tau_{\text{transverse}} \tag{17}$$

The principal stresses are:

$$\sigma_{1,2} = \frac{\sigma_x}{2} \pm \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau_{xy}^2} \tag{18}$$

The maximum shear stress is:

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau_{xy}^2} \tag{19}$$

See Figure 9 for the corresponding Mohr's circle.

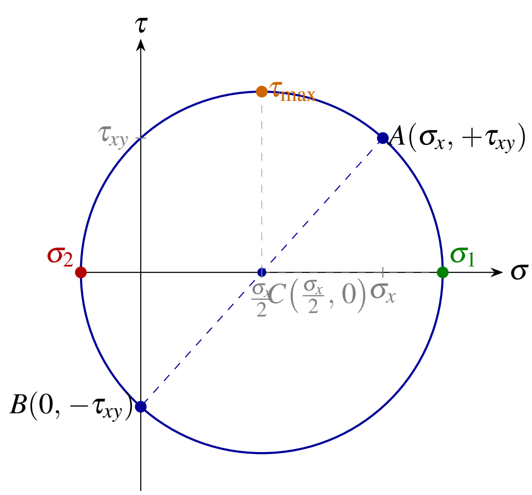


Figure 9: Mohr's circle — Beam B (Point F): combined bending and transverse shear.

Support Arms:

The support arms are subjected primarily to axial compression:

$$\sigma_1 = 0, \quad \sigma_2 = -\sigma_{\text{axial}} \tag{20}$$

$$\tau_{\max} = \frac{|\sigma_{\text{axial}}|}{2} \tag{21}$$

See Figure 10 for the corresponding Mohr's circle.

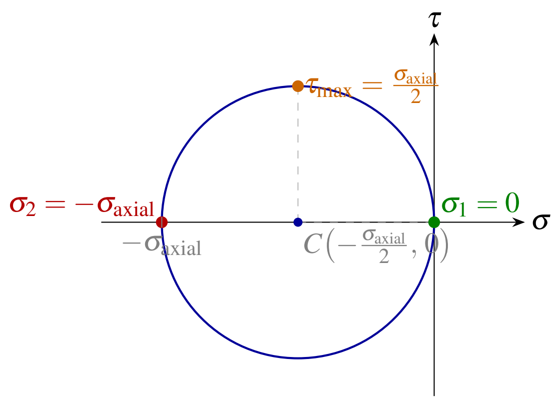


Figure 10: Mohr's circle — Support arms: uniaxial compression.

Pins (e.g., Pin F):

Pins experience pure shear:

$$\sigma_1 = \frac{\tau_{\text{direct}}}{2}, \quad \sigma_2 = -\frac{\tau_{\text{direct}}}{2} \tag{22}$$

$$\tau_{\max} = \tau_{\text{direct}} \tag{23}$$

See Figure 11 for the corresponding Mohr's circle.

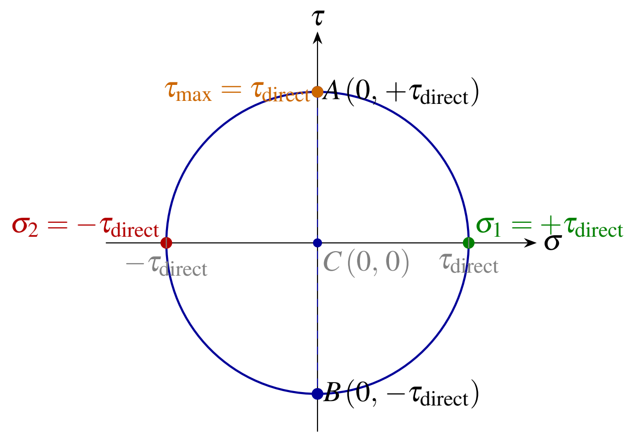


Figure 11: Mohr's circle — Pins (e.g. Pin F): pure shear.

Pump Plate:

The pump plate experiences combined bending and shear similar to the beam:

$$\sigma_{1,2} = \frac{\sigma_{bending}}{2} \pm \sqrt{\left(\frac{\sigma_{bending}}{2}\right)^2 + \tau_{transverse}^2} \tag{24}$$

$$\tau_{max} = \sqrt{\left(\frac{\sigma_{bending}}{2}\right)^2 + \tau_{transverse}^2} \tag{25}$$

See Figure 12 for the corresponding Mohr's circle.

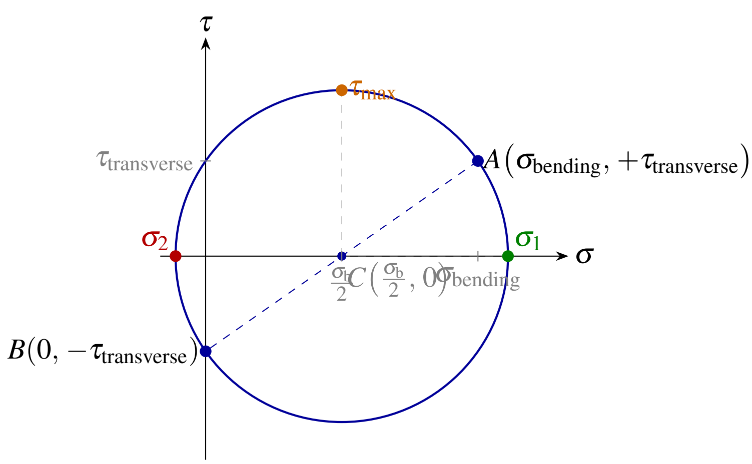


Figure 12: Mohr's circle — Pump plate: combined bending and transverse shear.

These principal and shear stress values are used in the following section to evaluate failure criteria and factor of safety. Detailed derivations are provided in Appendix A.

4. Failure Determination (Symbolic)

4.1 Factor of Safety Expressions

Failure of the structural components was evaluated using the Distortion Energy (Von Mises) criterion, which is appropriate for ductile materials under multi-axial loading. This criterion was selected because it provides an accurate prediction of yielding and is less conservative than maximum shear stress theory.

The equivalent Von Mises stress is given by:

$$\sigma'_{\text{DE}} = \sqrt{\sigma_a^2 - \sigma_a\sigma_b + \sigma_b^2} \tag{26}$$

The factor of safety is defined as:

$$n = \frac{S_y}{\sigma'_{\text{DE}}} \tag{27}$$

Beam B – Critical Location (Point F):

$$\sigma'_{\text{DE, beam}} = \sqrt{\sigma_{a,\text{beam}}^2 - \sigma_{a,\text{beam}}\sigma_{b,\text{beam}} + \sigma_{b,\text{beam}}^2} \tag{28}$$

$$n_{\text{beam}} = \frac{S_y}{\sigma'_{\text{DE, beam}}} \tag{29}$$

Support Arms:

The support arms are primarily under axial compression:

$$\sigma'_{\text{DE, arm}} = |\sigma_{\text{axial}}| \tag{30}$$

$$n_{\text{arm}} = \frac{S_y}{\sigma'_{\text{DE, arm}}} \tag{31}$$

Pins (e.g., Pin F):

For shear-dominated loading:

$$\sigma'_{\text{DE, pin}} = \sqrt{\sigma_{a,\text{pin}}^2 - \sigma_{a,\text{pin}}\sigma_{b,\text{pin}} + \sigma_{b,\text{pin}}^2} \tag{32}$$

$$n_{\text{pin}} = \frac{S_y}{\sigma'_{\text{DE, pin}}} \tag{33}$$

Pump Plate:

$$\sigma'_{\text{DE, plate}} = \sqrt{\sigma_{a,\text{plate}}^2 - \sigma_{a,\text{plate}}\sigma_{b,\text{plate}} + \sigma_{b,\text{plate}}^2} \tag{34}$$

$$n_{\text{plate}} = \frac{S_y}{\sigma'_{\text{DE, plate}}} \tag{35}$$

These expressions are evaluated numerically in Section 5 to verify that all components meet the required minimum factor of safety of 1.5. Expanded derivations are provided in Appendix A.

4.2 Maximum Deflection of Pump Platform

The maximum deflection of the pump platform was determined using energy methods, modeling the platform as a beam subjected to multiple applied loads. The method of virtual work was used to account for the distributed effects of loading along the platform.

The deflection at the end of the platform is given by:

$$\delta = \frac{1}{EI}\left[(P\sin\theta + F_1)\frac{L_1^3}{3} + F_2\left(\frac{L_1^3 - L_2^3}{3} - \frac{L_2(L_1^2 - L_2^2)}{2}\right)\right] \tag{36}$$

where E is the modulus of elasticity, I is the second moment of area of the platform cross-section, and L_1, L_2, F_1, and F_2 correspond to the geometric and loading parameters defined from the equilibrium analysis.

This expression captures the combined effect of the applied pump load and support reactions along the platform. The calculated deflection is compared against the design requirement of $\delta \leq 1$ mm.

A full derivation of this expression is provided in Appendix B.

4.3 Critical Buckling Load of Support Arms

The support arms are subjected primarily to axial compressive loading and are therefore susceptible to buckling. Euler buckling theory was used to determine the critical load at which instability occurs.

For a pin–pin column, the critical buckling load is given by:

$$P_{cr} = \frac{\pi^2 EI}{L^2} \tag{37}$$

where E is the modulus of elasticity, I is the second moment of area of the cross-section, and L is the effective length of the support arm.

The corresponding factor of safety against buckling is:

$$n_{\text{buckle}} = \frac{P_{cr}}{P_{\text{axial}}} = \frac{\pi^2 EI}{L^2 P_{\text{axial}}} \tag{38}$$

This expression is used to verify that the support arms meet the minimum required factor of safety of 1.5. A full derivation and geometric substitutions are provided in Appendix C.

5. Design Iteration

5.1 Initial Dimensions

The initial dimensions for each component were selected based on geometric constraints of the mounting frame and preliminary stress analysis. All dimensions are reported in inches.

Table 4: Initial design dimensions for all components.

Component	Variable	Initial Value	Units
Support arm length	L	13.8127	in
Support arm cross-section	$b \times h$	0.375 × 0.75	in
Beam length	L_b	14.7398	in
Beam outer dimensions	$b \times h$	0.375 × 0.375	in
Beam wall thickness	t	0.035	in
Pump platform dimensions	$L \times W$	4.1063 × 4.1314	in

Using the initial dimensions from Table 4, numerical values were substituted into the symbolic stress, deflection, and buckling expressions developed in Sections 3 and 4. The first iteration showed that the overall design was feasible, but the load path through the pump platform and beams could be improved.

To reduce bending effects and better distribute the load, the connection points between the beams and support arms were moved inward to approximately halfway along the pump platform. This change reduced the unsupported span of the platform and improved the structural behavior of the assembly by transferring load more effectively from the pump platform into the support arms.

The revised geometry was then evaluated using the same symbolic expressions for axial stress, bending stress, factor of safety, maximum deflection, and buckling load. With the updated connection locations, the design satisfied the required factor of safety, deflection, and buckling criteria. This iteration was therefore used as the basis for the final design and subsequent FEA validation.

5.2 First Design Iteration

Using the dimensions in Table 4, numerical values were substituted into the symbolic expressions developed in Sections 3 and 4 to evaluate stresses, deflection, and buckling.

Support Arm (Axial Stress):

$$A = bh = (0.375)(0.75) = 0.2813 \text{ in}^2 \tag{39}$$

$$\sigma_{\text{axial}} = \frac{P}{A} = \frac{P}{0.2813} \tag{40}$$

Buckling Check (Euler):

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(0.375)(0.75)^3 = 0.0132 \text{ in}^4 \tag{41}$$

$$P_{cr} = \frac{\pi^2 EI}{L^2} \tag{42}$$

Using $E = 10 \times 10^6$ psi (aluminum) and $L = 13.8127$ in:

$$P_{cr} = \frac{\pi^2(10 \times 10^6)(0.0132)}{(13.8127)^2} \approx 6.81 \times 10^3 \text{ lb} \tag{43}$$

$$n_{\text{buckle}} = \frac{P_{cr}}{P} \tag{44}$$

Beam (Bending Stress, Hollow Section):

Outer dimensions: $b_o = h_o = 0.375$ in, wall thickness $t = 0.035$ in Inner dimensions: $b_i = h_i = 0.375 - 2(0.035) = 0.305$ in

$$I = \frac{b_o h_o^3 - b_i h_i^3}{12} = \frac{(0.375)(0.375)^3 - (0.305)(0.305)^3}{12} \approx 8.89 \times 10^{-4} \text{ in}^4 \tag{45}$$

$$c = \frac{h_o}{2} = 0.1875 \text{ in} \tag{46}$$

$$\sigma_{\text{bending}} = \frac{Mc}{I} \tag{47}$$

Pump Platform Deflection:

From Section 4, the deflection is:

$$\delta = \frac{1.16 \times 10^8}{E} \tag{48}$$

Using $E = 10 \times 10^6$ psi:

$$\delta \approx 0.0116 \text{ in } (0.295 \text{ mm}) \tag{49}$$

This satisfies the requirement $\delta \leq 1$ mm.

Factor of Safety:

Using the Von Mises criterion:

$$n = \frac{S_y}{\sigma'_{\text{DE}}} \tag{50}$$

For aluminum ($S_y \approx 40{,}000$ psi) and the calculated stress values:

$$n > 1.5 \tag{51}$$

Summary:

The first design iteration satisfies all required design criteria:

- Factor of safety > 1.5
- Maximum deflection < 1 mm
- Buckling load exceeds applied load

Based on these results, the design was considered acceptable and used for further refinement and validation.

5.3 Ansys Material Selection Plots

Material selection was performed using Ashby charts in Ansys Granta EduPack to minimize mass while satisfying strength, stiffness, and stability requirements derived from the design analysis. The performance indices used for material selection were obtained from the governing stress, deflection, and buckling relationships for each component.

Constraints:

Before applying optimization functions, materials were filtered using the following constraints:

- Corrosion resistance due to operation near water
- Machinability with available tools
- Commercial availability (e.g., McMaster-Carr)
- Total material cost below \$150

Only materials satisfying all constraints were considered in the Ashby plots.

Performance Indices:

For the **beams and support arms**, which are governed by bending, axial loading, and buckling, the following performance indices were derived:

$$M_1 = \frac{S_y}{\rho} \quad \text{(strength-limited design)} \tag{52}$$

$$M_2 = \frac{E}{\rho} \quad \text{(stiffness and buckling-limited design)} \tag{53}$$

These were obtained by minimizing mass subject to bending stress and Euler buckling constraints. For the **pump platform**, which is governed by bending stress and deflection, the derived performance indices were:

$$M_3 = \frac{S_y^{1/2}}{\rho} \tag{54}$$

$$M_4 = \frac{E^{1/3}}{\rho} \tag{55}$$

These indices result from substituting geometric relationships into the bending stress and deflection equations and solving for minimum mass.

Ashby Plots:

Ashby plots of yield strength vs. density and Young's modulus vs. density were generated using these performance indices (see Figures 13 and 14). Materials located toward the upper-left region of these plots maximize the performance indices, corresponding to high strength or stiffness with low density.

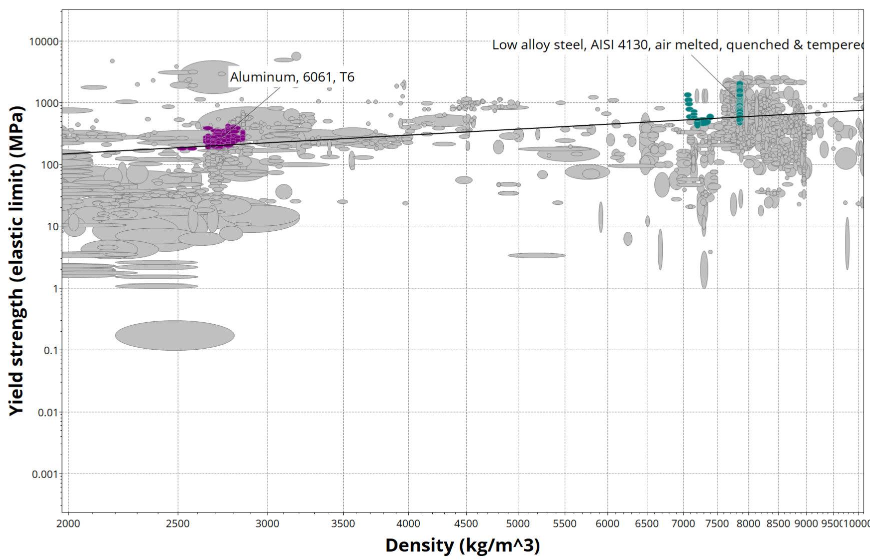


Figure 13: Ashby plot of yield strength vs. density used for strength-based material selection.

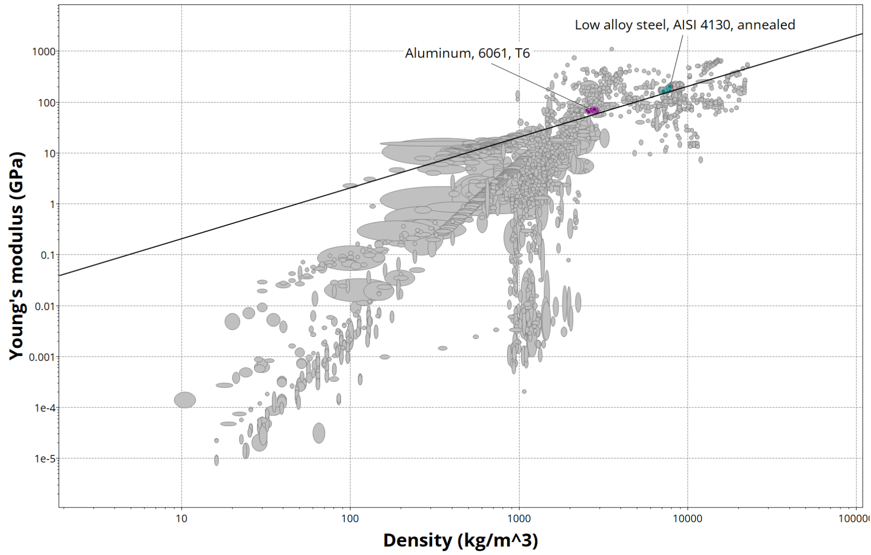


Figure 14: Ashby plot of Young's modulus vs. density used for stiffness and buckling considerations.

Material Selection Outcome:

From these plots, aluminum alloys and alloy steels were consistently identified as optimal candidates. Aluminum alloys provide significant weight reduction while maintaining adequate stiffness, while alloy steels offer higher strength where required. The final material choices reflect a balance between these properties while satisfying all design constraints.

5.4 Further Iterations

With candidate materials identified from the Ashby analysis, the symbolic stress, deflection, and buckling expressions from Sections 3 and 4 were re-evaluated using the updated material properties. 4140 alloy steel was assigned to the main beams due to its high yield strength and modulus, both of which directly improve bending resistance at the critical pin hole on Beam B and reduce the pump platform deflection. 6061-T6 aluminum was assigned to the support arms and the pump plate, where its lower density offered weight savings and its modulus was still sufficient to maintain an Euler buckling factor of safety well above 1.5 given the axial compressive load of 317.39 N carried by each support arm.

With the material assignments fixed, a further pass was made on the component cross-sections and connection geometry. The beam wall thickness was kept at 0.035 in to match readily available hollow stock from McMaster-Carr, while the support arm cross-section was retained at 0.375 × 0.75 in after the buckling check confirmed a large margin against instability. The pump plate thickness was verified against both the bending stress from the applied moment M_{Py} and the 1 mm deflection requirement, and the bolt pattern was kept consistent with the pump manufacturer's specified mounting geometry to avoid introducing additional machining operations. The support arm attachment location on the main beam was left at the mid-platform position established in Section 5.1, since moving it further outboard reduced bending stress in the beam but also increased the effective length of the support arm, reducing its buckling margin.

5.5 Final Dimensions and Materials

Table 5: Final design dimensions and selected materials.

Component	Final Dimension	Material	Cost ($)
Steel Arms (×2)	$L = 374.5$ mm, 9.5×9.5 mm hollow	4140 Alloy Steel	48.29
Support Arms (×4)	$L = 350.9$ mm, 9.5×6.4 mm	6061-T6 Aluminum	5.69
Pump Plate	$104.3 \times 104.9 \times 6.4$ mm	6061-T6 Aluminum	27.44
Spacers	$d = 9.5$ mm, $L = 11.1$ mm	6061-T6 Aluminum	2.94
Fasteners	10-32, 1.5 in and 0.625 in	High-Strength Steel	15.76
Total			**$100.12**

6. Finite Element Validation

6.1 FEA Setup

Our design was further validated by putting our 3D model into a FEA simulation. This FEA simulation was done using SolidWorks FEA software. To set up the simulation, the model was fixed at the end of the aluminum and steel arms where they meet the mounting holes at the wall. Then, 4 loads were placed at each of the bolt holes for where the pump mounts to the pump plate to account for the weight of the pump. Lastly, two remote loads were placed at the positions where the pump stand pulley was located to accurately account for all the forces and moments caused by that load on the pump stand assembly.

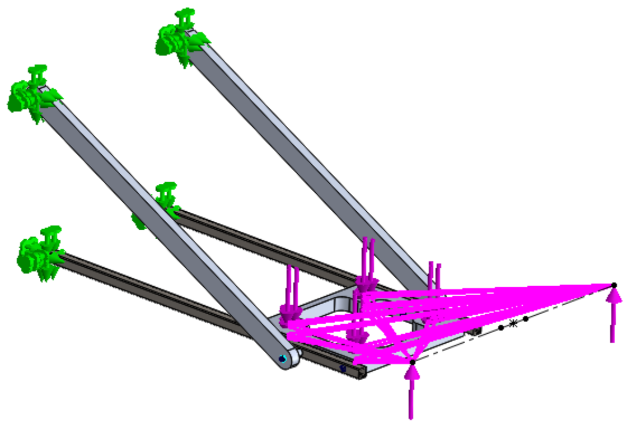


Figure 15: FEA assembly screenshot highlighting connection points, applied loads, and suppressed components.

6.2 CAD Assembly

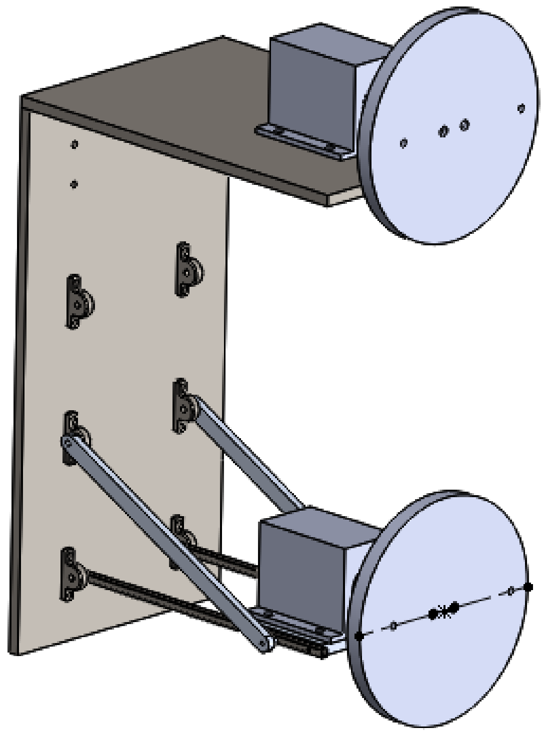


Figure 16: CAD assembly screenshot

The assembly configuration is shown in Fig. 16. To construct the assembly, the wall and mounting were both fixed. Then, both the steel arms were mated to the mounting holes on the wall. After that, the pump plate was mated in between the steel arms. Then, the spacers and aluminum support arms were mated to the steel arms and the mounting holes on the wall. Finally, both pumps and pulleys were mated to the pump stand and mounting frame respectively.

6.3 FEA Results

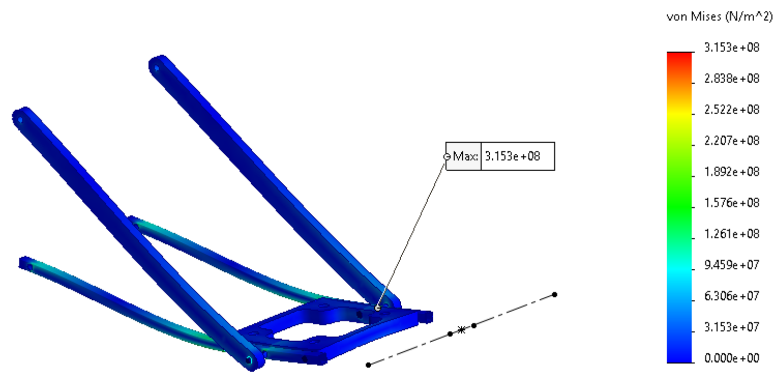


Figure 17: FEA stress results for the full assembly. Maximum stress location is indicated.

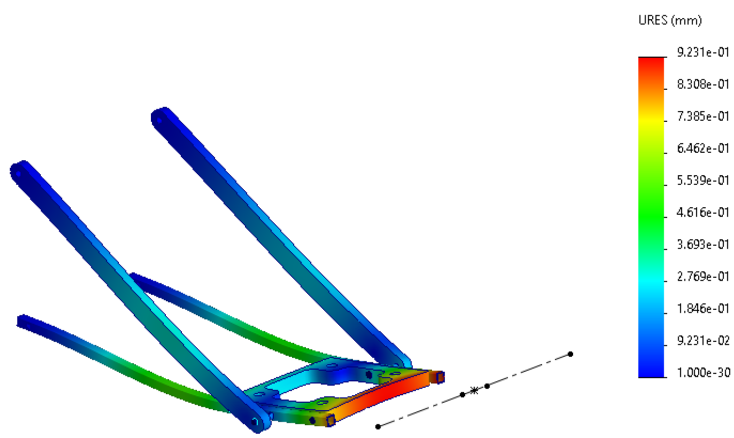


Figure 18: FEA deflection results for the full assembly.

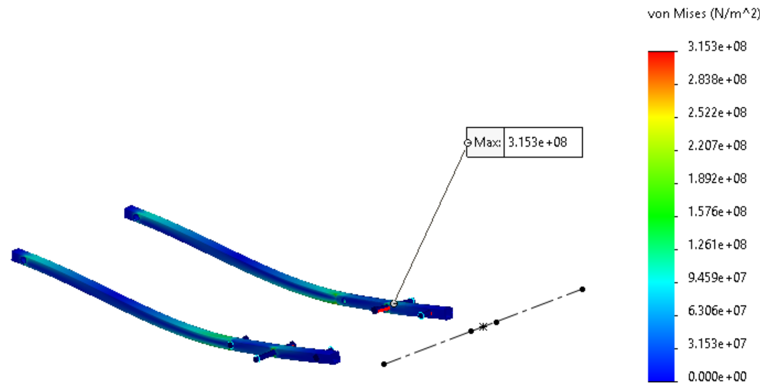


Figure 19: FEA stress results for the steel arms (individual component).

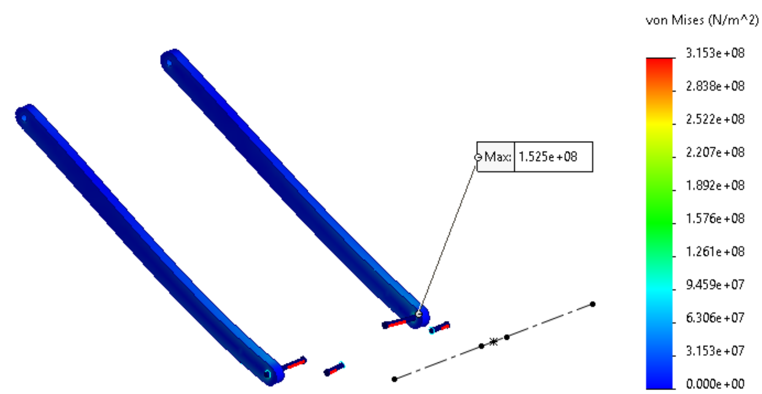


Figure 20: FEA stress results for the support arms (individual component).

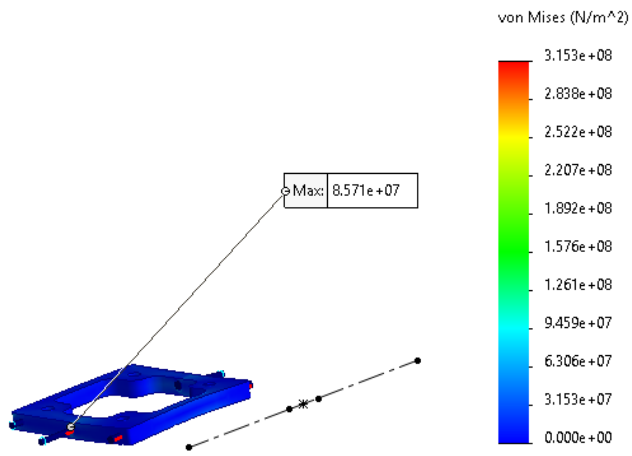


Figure 21: FEA stress results for the pump plate (individual component).

The full-assembly results are shown in Fig. 17 and Fig. 18. Individual component results are shown in Fig. 19, Fig. 20, and Fig. 21.

6.4 Convergence Study

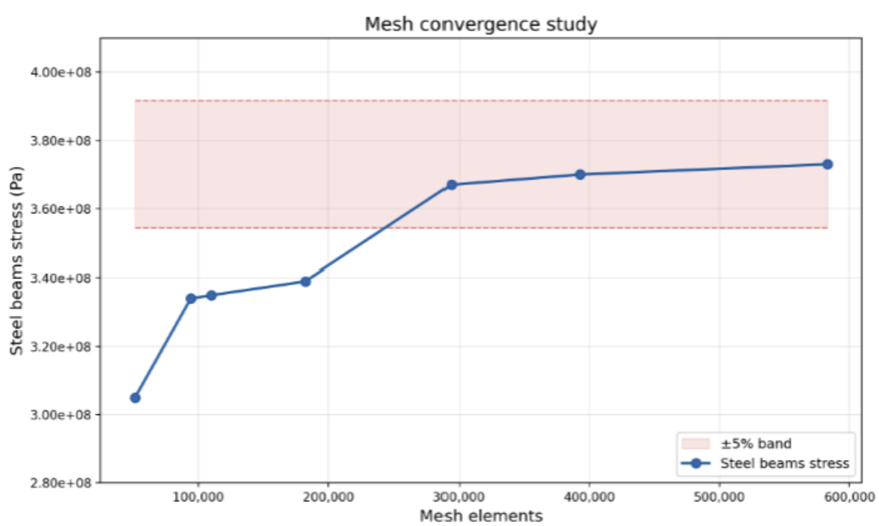


Figure 22: Mesh convergence plot showing maximum von Mises stress vs. number of mesh elements.

The convergence plot (Fig. 22) was created by running several different studies at differing mesh quality levels. During the study, we looked at the change in stress values for all the components, but it was the steel arms that had the most fluctuation, so those stress values were used in the convergence study to define convergence.

6.5 GA3 Questions

Comparison of Predicted Failure Locations:

Based on the FEA Von Mises stress results and corresponding material yield strengths, the component with the lowest factor of safety is the bolt at Point F, making it the most likely location for initial failure. The next most critical component is Beam B, with the highest stress occurring at the top edge of the pin hole due to stress concentration effects.

These results are consistent with the GA2 predictions, where Pin F was identified as the most critical component, followed by Beam B. This agreement indicates that the analytical stress analysis successfully captured the dominant loading conditions and critical regions of the design.

Highest Stress Locations:

From the GA2 analysis, the expected critical locations were the top edges of pin holes at points E and F for the beams, the midpoints of the support arms due to compressive loading, and the bolt hole regions of the pump platform.

FEA results confirmed that the highest stresses in the beams occurred at the top of the holes at points E and F. For the support arms, however, the highest stresses were observed at the connection points to the beams rather than at the midpoints. Similarly, the pump platform exhibited maximum stress at its connection to Beam B rather than at the pump mounting holes.

These differences are likely due to the ability of FEA to capture localized stress concentrations and connection effects that were not fully represented in the simplified analytical models.

Platform Deflection:

The FEA results predicted a maximum platform deflection of approximately 0.9 mm at the farthest point from the wall. The analytical result from GA2 was:

$$\delta = \frac{1.16 \times 10^8}{E} \tag{56}$$

For 4140 alloy steel ($E \approx 210$ GPa), this yields a deflection of approximately 0.56 mm. The higher deflection predicted by FEA is expected, as the analytical model assumes simplified beam behavior, while FEA accounts for full 3-D geometry, connection compliance, and stress concentrations.

Design Iteration Insights:

The FEA results suggest that stress concentrations at points E and F dominate the structural response. Further design optimization could involve adjusting the positions of these connection points to reduce peak stresses and improve load distribution.

However, the current design already reflects iterative improvements and satisfies the required performance criteria. Additional optimization would likely yield diminishing returns relative to the scope of the project.

7. Design Changes

Using our FEA model, we attempted and made some different design changes to try and make our pump stand perform better. For example, we moved the mounting location for the support arms to be further up the beam, as this decreased the overall deflection in the beam. We also attempted varying the thickness of our support arms to see if thinner support arms would be able to support the given loads.

8. CAD and Fabrication Drawings

8.1 CAD Drawings

Below are the CAD drawings that were used to fabricate the different parts in our pump stand assembly.

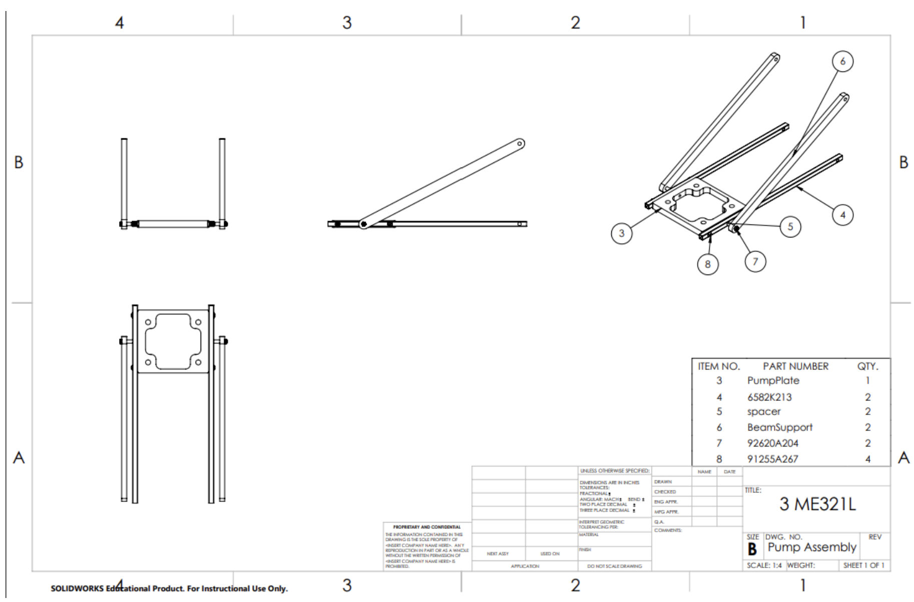


Figure 23: CAD fabrication drawing of the steel arms. All dimensions in mm.

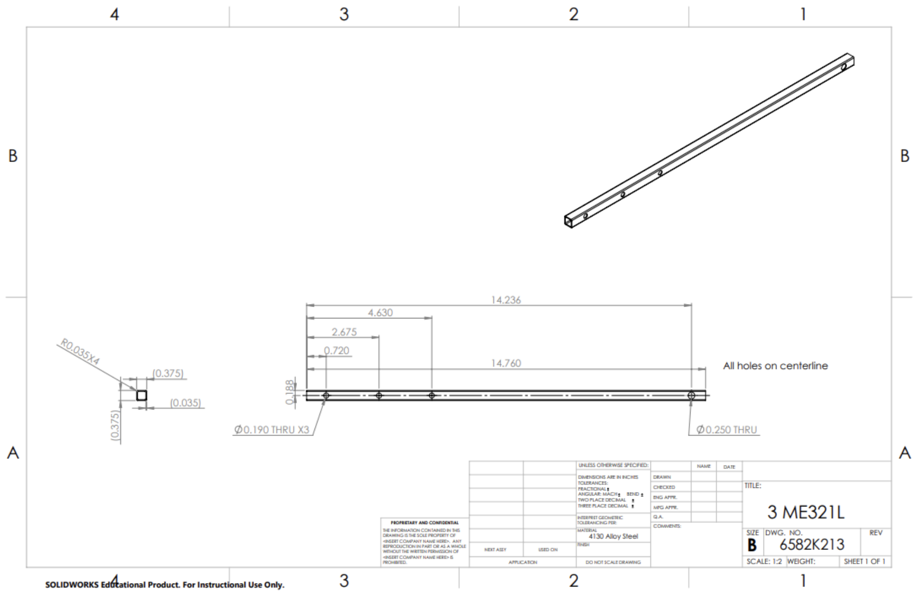


Figure 24: CAD fabrication drawing of the steel arms. All dimensions in mm.

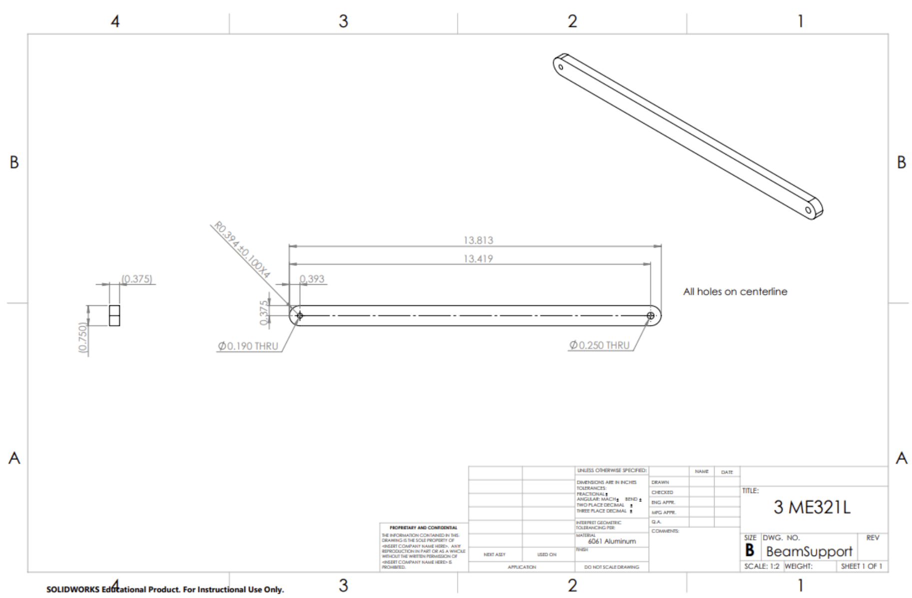


Figure 25: CAD fabrication drawing of Support Arms. All dimensions in mm.

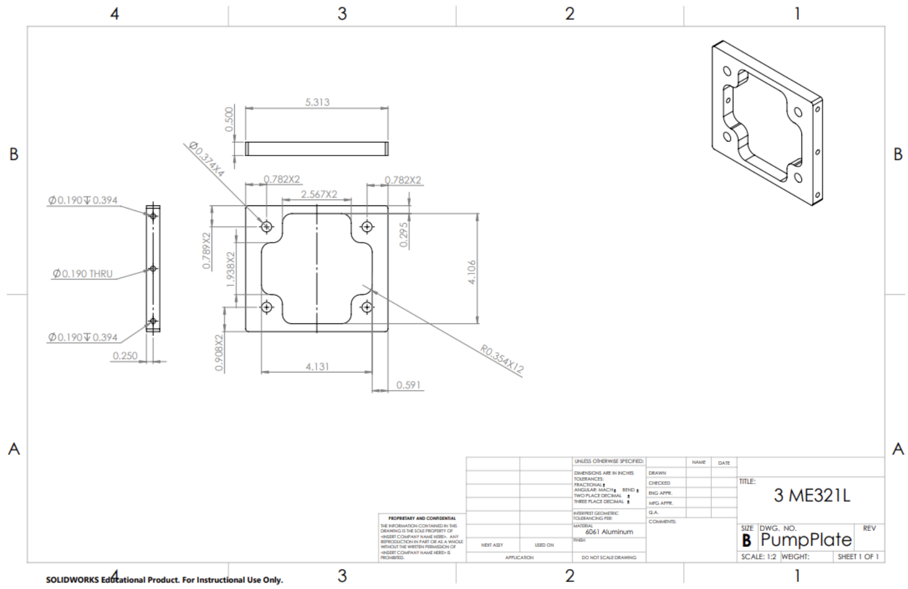


Figure 26: CAD fabrication drawing of the pump plate. All dimensions in mm.

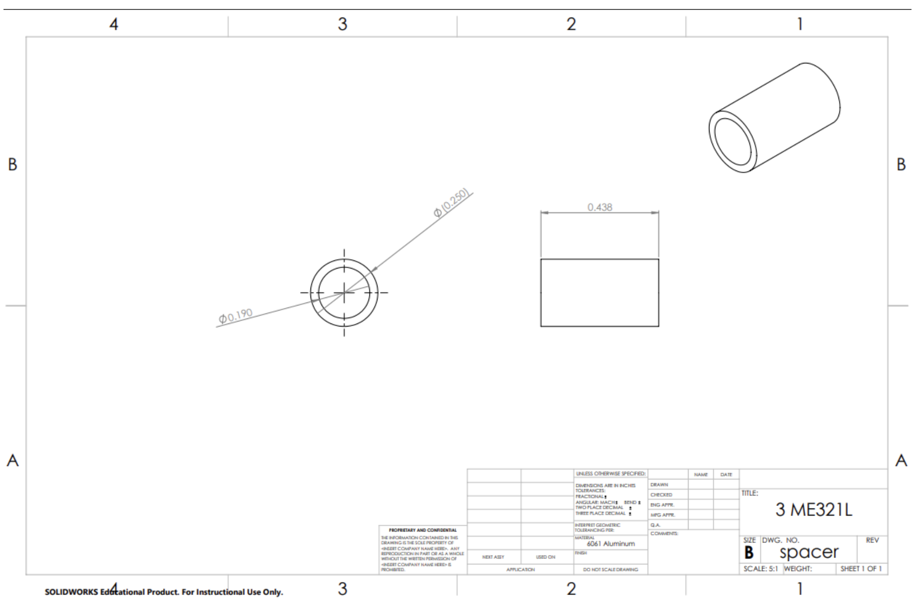


Figure 27: CAD fabrication drawing of support arm spacers. All dimensions in mm.

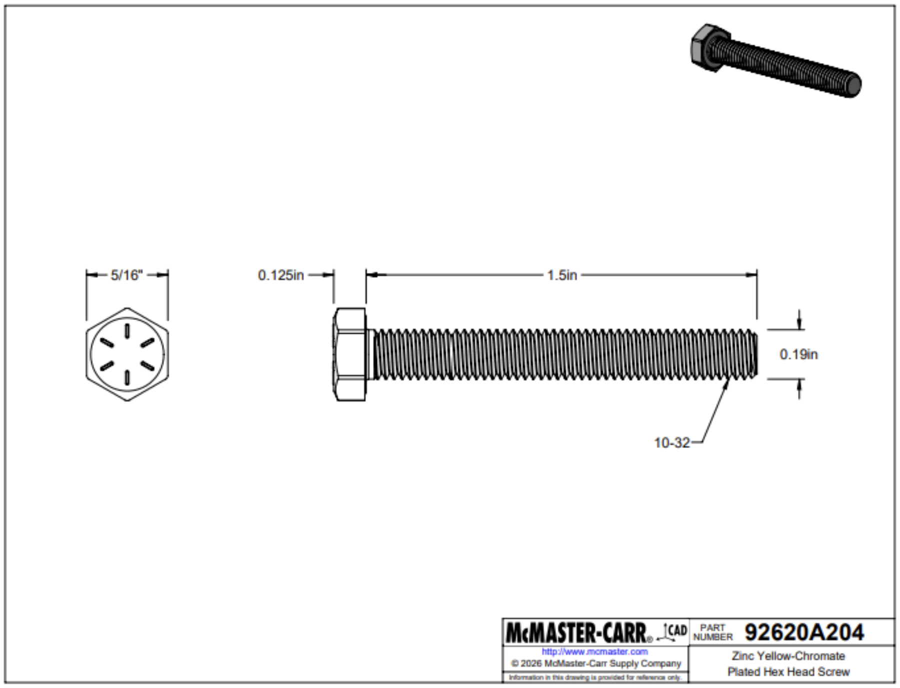


Figure 28: CAD fabrication drawing of bolts. All dimensions in mm.

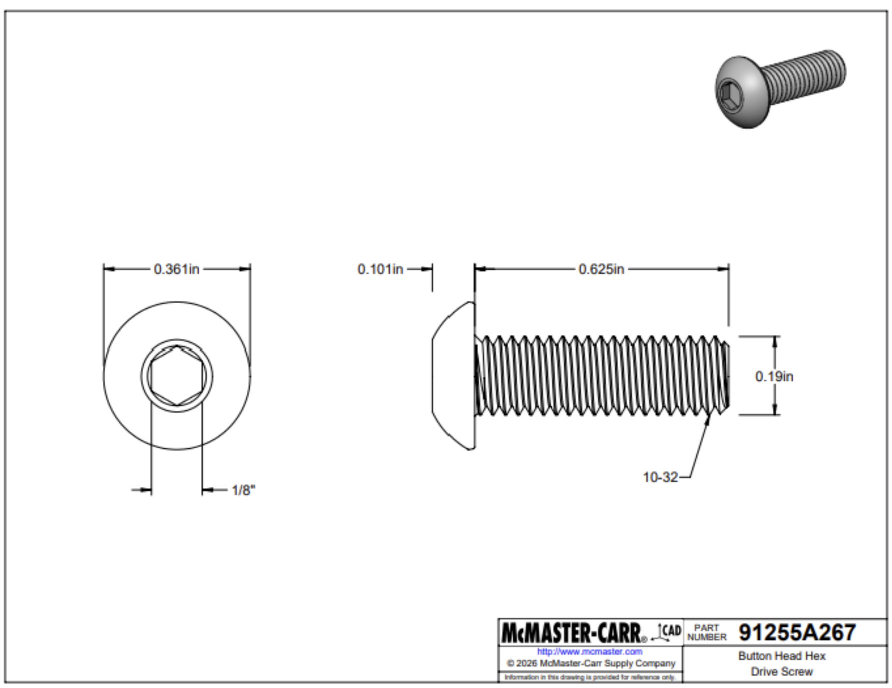


Figure 29: CAD fabrication drawing of bolts. All dimensions in mm.

Drawings are shown in Fig. 23, Fig. 24, Fig. 25, Fig. 26, Fig. 27, Fig. 28, and Fig. 29.

8.2 Bill of Materials

Table 6: Bill of materials for the final design.

Item	Description	Qty	Material	Unit Cost ($)	Total ($)
1	Shaft	2	6061 Aluminum	2.94	5.88
2	Arm	2	4140 Steel	48.29	48.29
3	Mounting plate	1	6061 Aluminum	27.44	27.44
4	Bolts (10-32)	4	High Strength Steel	3.77	15.76
5	Support Arms	2	6061 Aluminum	5.69	11.38
	Total				**$108.75**

8.3 Physical Component Photographs

Figure 30: Photographs of the fabricated individual components prior to assembly.

9. Physical Prototype Performance Results

Table 7: Performance results recorded by instructor/TA during GA5 testing.

Metric	Result	Units
Weight	0.818	kg
Connected properly	Yes	–
Installation/assembly time	1:45	–
Height	15.1	cm
Maximum load	20.6	kg
Maximum deflection	4.9	mm

Figure 31: Photograph of the physical prototype installed in the testing frame during GA5.

10. Discussion

(i) Interface with testing setup:

The design successfully interfaced with the testing setup and was installed without modification. All components aligned as intended and the system functioned as designed.

(ii) Installation speed:

The installation time of 1:45 was efficient and met expectations. The assembly process was straightforward, with no major delays or complications during setup.

(iii) Withstanding maximum applied load:

The prototype successfully withstood the applied load without failure, indicating that the design and material selection met the required strength criteria.

(iv) Deflection criteria:

The design did not meet the deflection requirement of less than 1 mm, with a measured deflection of 4.9 mm. This was likely not due solely to elastic deformation of the members, but also due to compliance in the pin connections. The weight of the pump introduced an initial downward displacement, and when the upward load was applied during testing, small clearances in the joints allowed rotation and "take-up" of slack. This resulted in additional apparent deflection beyond what was predicted analytically.

(v) Comparison of calculated FOS, FEA, and physical performance:

The calculated factor of safety and FEA results predicted acceptable stresses and minimal deflection; however, the physical prototype exhibited significantly greater deflection. This discrepancy is likely due to idealized assumptions in the models, particularly treating connections as rigid. In reality, joint clearances and load reversal (from the pump weight to applied load) introduced additional displacement not captured in the analysis.

(vi) Unexpected challenges:

The primary challenge was unintended compliance in the pin connections. Small clearances allowed slight movement in the support arms, which was not significant under the initial pump weight but became important when the loading direction reversed during testing. This resulted in additional displacement due to joint rotation and slack take-up. This highlights the importance of accounting for connection behavior and tolerances in addition to member strength in future designs.

11. Conclusion

This project involved designing, analyzing, and fabricating a structural support frame for a belt-driven centrifugal pump. Over the course of the semester, the work moved from symbolic equilibrium and stress analysis to numerical design iteration, FEA validation, and physical prototype testing. The design process began with equilibrium and stress analyses to identify internal forces, critical locations, and governing failure modes, which were used to develop symbolic expressions for stress, deflection, and factor of safety. These expressions were then evaluated during iterative design refinement and validated against both computational and experimental results.

Several engineering tools were central to the analysis. Equilibrium was solved symbolically using SymPy, which made design iteration straightforward since dimensions could be updated without re-deriving anything. The Von Mises criterion was used for failure prediction, appropriate for the ductile materials in the design. Deflection was calculated using virtual work, and Euler buckling theory was applied to check the aluminum support arms. Material selection relied on Ashby charts in Ansys Granta EduPack, using performance indices developed from the governing stress and deflection equations to find materials that balanced strength, stiffness, weight, and cost. The final structure used 4140 alloy steel for the main beams and 6061-T6 aluminum for the support arms and pump plate, providing adequate corrosion resistance and machinability while keeping cost and weight low.

The prototype met the majority of the imposed design requirements. A total material cost of \$108.75 remained well under the \$150 budget, and the fabricated weight of 0.818 kg reflected the material selection process working as intended. The structure interfaced with the testing frame without modification and installed in 1 minute and 45 seconds with no fit issues, which points to accurate fabrication and a design that was practical to assemble. It also withstood the maximum applied load of 20.6 kg without yielding or buckling, confirming that the strength-based factor of safety criterion was satisfied. FEA results agreed closely with the analytical predictions of critical stress locations, with Pin F and the pin hole at Point F on Beam B identified as the most highly stressed regions in both analyses.

The deflection requirement of less than 1 mm was not met, with a measured value of 4.9 mm. Both the analytical model (0.56 mm) and FEA (0.9 mm) predicted compliance with the requirement, so the gap between prediction and test result was substantial. The most likely cause is pin joint compliance. Small clearances in the connections allowed rotation when the load direction reversed from pump weight to applied test load, adding displacement that neither model accounted for, as both treated the connections as rigid. Addressing this in a future design would mean tighter pin tolerances, bushings or press-fit pins to reduce clearance, or a different connection approach entirely. The broader takeaway is that joint behavior needs to be part of the analysis, not just member stiffness.

Overall, the core design objectives around strength, cost, and usability were achieved, and the deflection issue is a well-understood problem with a clear path to fixing it. The project demonstrated the value of combining analytical, computational, and experimental methods in mechanical design and highlighted the importance of accounting for connection behavior alongside member-level strength and stiffness in structural assemblies.

A. Detailed Stress Derivations

The following expressions show the full derivation of stresses used in the analysis, including intermediate steps and substitutions based on geometry.

A.1 Beam B – Point F

Bending Stress Derivation:

$$M = \frac{PL_{\text{beam}}}{2} \tag{57}$$

$$c = \frac{W_o}{2} \tag{58}$$

$$I = \frac{W_o^4 - w_i^4}{12} \tag{59}$$

$$\begin{aligned} \sigma_{\text{bending}} &= \frac{Mc}{I} && (60) \\ &= \frac{\left(\frac{PL_{\text{beam}}}{2}\right)\left(\frac{W_o}{2}\right)}{\frac{W_o^4 - w_i^4}{12}} && (61) \\ &= \frac{PL_{\text{beam}} W_o}{4} \cdot \frac{12}{W_o^4 - w_i^4} && (62) \\ &= \frac{3PL_{\text{beam}} W_o}{W_o^4 - w_i^4} && (63) \end{aligned}$$

Axial Stress:

$$A = w_{\text{beam}} h_{\text{beam}}, \quad \sigma_{\text{axial}} = \frac{P}{w_{\text{beam}} h_{\text{beam}}} \tag{64}$$

Transverse Shear:

$$\tau = \frac{2V}{A} = \frac{2P}{W_o^2} \tag{65}$$

A.2 Principal Stress Calculation

$$\sigma_a = \frac{\sigma}{2} + \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} \tag{66}$$

$$\sigma_b = \frac{\sigma}{2} - \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} \tag{67}$$

A.3 Pump Plate Deflection

$$\delta = \frac{1}{EI}\left[\int_0^{L_2} (P\sin\theta + F_1)x^2 dx + \int_{L_2}^{L_1} \left((P\sin\theta + F_1)x^2 + F_2 x(x - L_2)\right) dx\right] \tag{68}$$

$$\delta = \frac{1}{EI}\left[(P\sin\theta + F_1)\frac{L_1^3}{3} + F_2\left(\frac{L_1^3 - L_2^3}{3} - \frac{L_2(L_1^2 - L_2^2)}{2}\right)\right] \tag{69}$$

A.4 Expanded Principal Stress Calculations

Beam B – Full Expansion:

$$\sigma_x = \frac{3PL_{\text{beam}}W_o}{W_o^4 - w_i^4} \tag{70}$$

$$\tau_{xy} = \frac{2P}{W_o^2} \tag{71}$$

$$\sigma_{1,2} = \frac{\sigma_x}{2} \pm \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau_{xy}^2} \tag{72}$$

Substituting:

$$\sigma_{1,2} = \frac{1}{2}\left(\frac{3PL_{\text{beam}}W_o}{W_o^4 - w_i^4}\right) \pm \sqrt{\left(\frac{3PL_{\text{beam}}W_o}{2(W_o^4 - w_i^4)}\right)^2 + \left(\frac{2P}{W_o^2}\right)^2} \tag{73}$$

Maximum Shear:

$$\tau_{\max} = \sqrt{\left(\frac{3PL_{\text{beam}}W_o}{2(W_o^4 - w_i^4)}\right)^2 + \left(\frac{2P}{W_o^2}\right)^2} \tag{74}$$

A.5 Pump Plate Expansion

$$\sigma_x = \frac{3P_pL_p}{W_ph_p^2} \tag{75}$$

$$\tau_{xy} = \frac{2P}{W_ph_p} \tag{76}$$

$$\sigma_{1,2} = \frac{\sigma_x}{2} \pm \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau_{xy}^2} \tag{77}$$

A.6 Pin Stress Expansion

$$\tau_{\text{direct}} = \frac{4P}{\pi d^2} \tag{78}$$

$$\sigma_{1,2} = \pm\frac{\tau_{\text{direct}}}{2} \tag{79}$$

B. Detailed Deflection Derivation

The maximum deflection of the pump platform was determined using the method of virtual work. The platform is modeled as a beam with multiple applied forces, and the deflection is computed by integrating the product of the moment distribution and a virtual unit load.

Moment Derivatives:

$$\frac{\partial M_1}{\partial Q} = -x \tag{80}$$

$$\frac{\partial M_2}{\partial Q} = -x \tag{81}$$

Deflection Expression:

$$\delta = \frac{1}{EI}\left[\int_0^{L_2} M_1(x)\frac{\partial M_1}{\partial Q}dx + \int_{L_2}^{L_1} M_2(x)\frac{\partial M_2}{\partial Q}dx\right] \tag{82}$$

Substituting Moment Expressions:

$$\delta = \frac{1}{EI}\left[\int_0^{L_2}\big[-(P\sin\theta + F_1)x\big](-x)\,dx + \int_{L_2}^{L_1}\big[-(P\sin\theta + F_1)x - F_2(x-L_2)\big](-x)\,dx\right] \tag{83}$$

Simplified Form:

$$\delta = \frac{1}{EI}\left[\int_0^{L_2}(P\sin\theta + F_1)x^2\,dx + \int_{L_2}^{L_1}\big[(P\sin\theta + F_1)x^2 + F_2x(x-L_2)\big]\,dx\right] \tag{84}$$

Evaluating the Integrals:

$$I_1 = (P\sin\theta + F_1)\int_0^{L_2} x^2dx = (P\sin\theta + F_1)\frac{L_2^3}{3} \tag{85}$$

$$I_2 = (P\sin\theta + F_1)\int_{L_2}^{L_1} x^2dx + F_2\int_{L_2}^{L_1} x(x-L_2)dx \tag{86}$$

$$= (P\sin\theta + F_1)\left[\frac{x^3}{3}\right]_{L_2}^{L_1} + F_2\left[\frac{x^3}{3} - \frac{L_2x^2}{2}\right]_{L_2}^{L_1} \tag{87}$$

$$= (P\sin\theta + F_1)\frac{L_1^3 - L_2^3}{3} + F_2\left(\frac{L_1^3 - L_2^3}{3} - \frac{L_2(L_1^2 - L_2^2)}{2}\right) \tag{88}$$

Final Deflection Expression:

$$\delta = \frac{1}{EI}\left[(P\sin\theta + F_1)\frac{L_1^3}{3} + F_2\left(\frac{L_1^3 - L_2^3}{3} - \frac{L_2(L_1^2 - L_2^2)}{2}\right)\right] \tag{89}$$

C. Detailed Buckling Analysis

The support arms are modeled as slender columns subjected to axial compression. Euler buckling theory is used to determine the critical load at which the column becomes unstable.

Column Properties:

$$I = \frac{1}{12}bh^3 \tag{90}$$

where b and h are the cross-sectional dimensions of the support arm.

Critical Buckling Load:

For a pin–pin column ($K = 1$):

$$P_{cr} = \frac{C\pi^2 EI}{L^2} \tag{91}$$

$$P_{cr} = \frac{\pi^2 E}{L^2} \cdot \frac{1}{12}bh^3 \tag{92}$$

$$P_{cr} = \frac{\pi^2 Ebh^3}{12L^2} \tag{93}$$

Factor of Safety:

$$n_{\text{buckle}} = \frac{P_{cr}}{P_{\text{axial}}} \tag{94}$$

$$n_{\text{buckle}} = \frac{\pi^2 E b h^3}{12 L^2 P_{\text{axial}}} \tag{95}$$